1. All the memory compilers are located here

```
[jjiangan@ecelvd811 C907_eval_rtl_ust-hk_20251010]$ cd /dfs/app/tsmc_icdc/tsmc028/28HPCplus_RF/SRAM/Compiler
tsmc_n28hpcpmc_20120200_110a  tsn28hpcp2prf_20120200_200b        tsn28hpcpdpsram_20120200_130a  tsn28hpcpuhddpsram_20120200_170a
tsn28hpcp1prf_20120200_130a   tsn28hpcpd127spsram_20120200_180a  tsn28hpcprom_20120200_130a     tsn28hpcpuhdspsram_20120200_170a
[jjiangan@ecelvd811 Compiler]$ pwd
/dfs/app/tsmc_icdc/tsmc028/28HPCplus_RF/SRAM/Compiler
```

Each of the compiler generates one type of memory

Type	Compiler Name	Foundry Name	Prefix of Instance	Bitcell (um^2)	Description
1PRF	tsn28hpcp1prf	ts	5n	0.155	One Port Register File
2PRF	tsn28hpcp2prf	ts	6n	0.24	Two Port Register File
SPSRAM	tsn28hpcpd127spsram	ts	1n	0.127	High Density Single Port SRAM
UHDSPSRAM	tsn28hpcpuhdspsram	ts	1n	0.127	Ultra High Density Single Port SRAM
DPSRAM	tsn28hpcpdpsram	ts	dn	0.315	Dual Port SRAM
UHDDPSRAM	tsn28hpcpuhddpsram	ts	dn	0.127	Ultra High Density Dual Port SRAM
ROM	tsn28hpcprom	ts	3n	0.044	ROM
L1_Cache	temn28hpcphssrammacros	tem	5n	0.155	L1 Cache
L2_Cache	tsn28hpcpl2spsrammacros	ts	1n	0.127	L2 Cache

2. **.pl** takes an **config.txt** as input, and generates configuration, then .mco generates the **.lib** file you need.

```
[jjiangan@ecelvd811 Compiler]$ cd tsn28hpcpuhdspsram_20120200_170a
config.txt  README  tsn28hpcpuhdspsram_170a.pl  tsn28hpcpuhdspsram_20120200_170a.mco
```

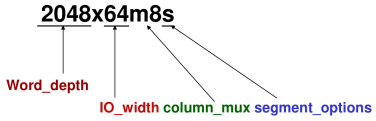
3. All the required memory lists can be found here

```
[jjiangan@ecelvd811 config]$ cd /dfs/usrhome/jjiangan/github/npu-charles/C907_eval_rtl_ust-hk_20251010/rtl/CPU
C907_asic.filelist  C907_asic_syn.filelist  INST_LIB  LIB  RTL
```

```
../../rtl/CPU/LIB/TS1N28HPCPUHDSVTB2048X112M4SWSO.v
../../rtl/CPU/LIB/TS1N28HPCPUHDSVTB2048X88M4SWSO.v
../../rtl/CPU/LIB/TS1N28HPCPUHDSVTB512X56M4SWSO.v
../../rtl/CPU/LIB/TS1N28HPCPUHDSVTB512X96M4SWSO.v
../../rtl/CPU/LIB/TS1N28HPCPHVTB4096X64M8SSO.v
../../rtl/CPU/LIB/TS1N28HPCPLVTB1024X68M8S.v
../../rtl/CPU/LIB/TS5N28HPCPLVTA256X58M4FW.v
../../rtl/CPU/LIB/TS5N28HPCPLVTA512X32M4FW.v
../../rtl/CPU/LIB/TS1N28HPCPSVTB1024X64M8SW.v
../../rtl/CPU/LIB/TS5N28HPCPSVTA64X124M2FW.v
../../rtl/CPU/LIB/TS1N28HPCPUHDSVTB64X24M2SW.v
../../rtl/CPU/LIB/TS1N28HPCPUHDSVTB256X118M2SW.v
../../rtl/CPU/LIB/TS1N28HPCPUHDSVTB256X78M2SW.v
../../rtl/CPU/LIB/DCCKBD16BWP30P140LVT.v
../../rtl/CPU/LIB/CKMUX2D4BWP30P140LVT.v
../../rtl/CPU/LIB/MUX2D4BWP30P140LVT.v
../../rtl/CPU/LIB/CKLNQD8BWP30P140LVT.v
```

4. Run the **gen_sram.sh** script here to generate the configurations. The input config is also provided.

   Note that in config.txt, TSMC only supports:



   The other options should be passed as flags to .pl file. So I suggest to input the **.pl** script to **Claude** and ask him how to generate the config.txt and command line script for generating SRAM macro **ts1n28hpcpuhdsvtm4swso** for you.

```
[jjiangan@ecelvd811 gen_sram]$ cd config/
ts1n_28hpcp_hvt_b4096x64_m8s_so.txt  ts1n_28hpcp_svt_b1024x64_m8s_w.txt  ts1n_28hpcp_uhd_svt_m4s_wso.txt  ts5n_28hpcp_svt_a64x124_m2f_w.txt
ts1n_28hpcp_lvt_b1024x68_m8s.txt     ts1n_28hpcp_uhd_svt_m2s_w.txt       ts5n_28hpcp_lvt_a_m4f_w.txt
```

```
[jjiangan@ecelvd811 tsn28hpcpuhdspsram_20120200_170a]$ cd /dfs/usrhome/jjiangan/github/npu-charles/C907_eval_rtl_ust-hk_20251010/smart_run/impl/gen_sram/
config  cvrt_lib2db.sh  db  gen_sram.sh
```

dc.tcl M | run_power_timebased_vcd.tcl U | gen_sram.sh | run_dc.csh M | tim_gatevcd.tcl | tim_gatevcd.tcl ↔ tim

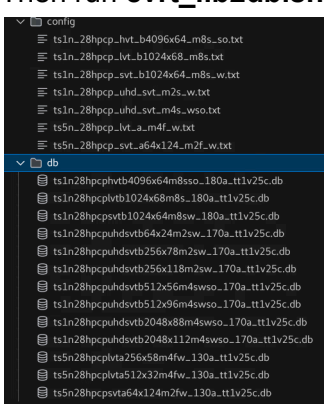
smart_run > impl > gen_sram > gen_sram.sh

```
#!/bin/tcsh
# Generate all SRAM macros required by C907 (ad_mp_top) using TSMC mc-2 compilers.
# Usage: source gen_sram.sh

# ---------------------------------------------------------------
# preparation
# ---------------------------------------------------------------
# clean workspace
rm -rf ts1*1*0a ts5*1*0a

# source the mc tool
source /dfs/app/tsmc_icdc/tsmc028/28HPCplus_RF/SRAM/Compiler/tsmc_n28hpcpmc_20120200_110a/cshrc.mc2

# ---------------------------------------------------------------
# generate:
#   ts1n28hpcphvtb4096x64m8sso
#   ts1n28hpcpsvtb1024x64m8sw
#   ts1n28hpcplvtb1024x68m8s
# ---------------------------------------------------------------
setenv MC_HOME /dfs/app/tsmc_icdc/tsmc028/28HPCplus_RF/SRAM/Compiler/tsn28hpcpd127spsram_20120200_180a
${MC_HOME}/tsn28hpcpd127spsram_180a.pl -file config/ts1n_28hpcp_hvt_b4096x64_m8s_so.txt \
    -NonBWEB -NonBIST -NonAWT -NLDM
```

Note: in the above script, you should source cshrc.mc2 first, then must set **MC_HOME** variable, and make it point to your compiler's path. Then after running .pl, .pl will automatically call .mco to do the rest of jobs

5. Then run **cvrt_lib2db.sh** to convert the lib to db. Finally you'll get